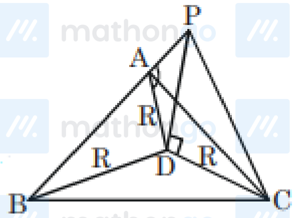
Q1 (2)

Let $\mathrm{PD}=\mathrm{h}, \mathrm{R}=2$

As angle of elevation of top of pole from $\mathrm{A}, \mathrm{B}, \mathrm{C}$ are equal

So D must be circumcentre of $\Delta \mathrm{ABC}$



$\tan \left(\frac{\pi}{3}\right)=\frac{\mathrm{PD}}{\mathrm{R}}=\frac{\mathrm{h}}{\mathrm{R}}$

$\mathrm{h}=\mathrm{R} \tan \left(\frac{\pi}{3}\right)=2 \sqrt{3}$

Q1: 18 March (Shift 2) - Single Correct

A pole stands vertically inside a triangular park ABC. Let the angle of elevation of the top of the pole from each corner of the park be $\frac{\pi}{3}$. If the radius of the circumcircle ot ΔABC is 2 , then the height of the pole is equal to:

(1) $\frac{2\sqrt{3}}{3}$

(2) $2\sqrt{3}$

(3) $\sqrt{3}$

(4) $\frac{1}{\sqrt{3}}$

Answer Key

Q1 (2)